

An effective finiteness theorem for an exceptional set of Erdős and Selfridge

Haoyu Chen
Independent researcher

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Abstract

For integers $n \geq 0$ and $k \geq 1$ let $v(n, k)$ be the number of prime factors of $n + k$ that divide none of $n, n + 1, \dots, n + k - 1$, equivalently the number of primes $p > k$ dividing $n + k$, and let $v_1(n) = \sup_{k \geq 1} v(n, k)$. Erdős and Selfridge (1967) expected $v_1(n) = 1$ to have only finitely many solutions but could not prove it. We prove that every $n \geq N_0$, where $\log N_0 = e^{45.28} \approx 4.62 \cdot 10^{19}$, has some $k \leq 10 \log n$ with $v(n, k) \geq 2$. Hence every solution of $v_1(n) = 1$ is smaller than N_0 . The proof combines an elementary counting lemma with Matveev's explicit lower bound for linear forms in logarithms; every numerical inequality is certified by interval arithmetic. The same argument gives, for all integers $l \geq 1$ and n with $\log \log n \geq 48.82$ and $\log n \geq l$, some $k \in [l, 10l \log n]$ with $v(n, k) \geq 2$. The qualitative finiteness also follows from a 1981 theorem of Langevin combined with an elementary lemma.

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1 Introduction

Erdős and Selfridge [ErSe67] considered, for integers $n \geq 0$ and $k \geq 0$, the number $v(n; k)$ of prime factors of $n + k$ that divide none of $n, n + 1, \dots, n + k - 1$, and the quantities

$$v_0(n) = \max_{k \geq 0} v(n; k), \quad v_l(n) = \max_{k \geq l} v(n; k).$$

They showed that $v_0(n) > 1$ for all n except $n = 1, 2, 3, 4, 7, 8, 16$, and wrote: “It seems certain that $\lim v_l(n) = \infty$ for every l , but unfortunately we have not even been able to prove that $v_1(n) = 1$ has only a finite number of solutions, though this certainly must be true. Probably the greatest n for which $v_1(n) = 1$ is $n = 330$.” They listed the 29 values $n < 2500$ with $v_1(n) = 1$: $n = 1\text{--}4, 6\text{--}8, 10, 12, 15, 16, 18, 22, 24, 26, 30, 36, 42, 46, 48, 60, 70, 78, 80, 96, 120, 190, 222, 330$. Guy's *Unsolved Problems in Number Theory* [Gu04, B27] records the question in its third edition (2004), repeating that Erdős and Selfridge were unable to prove that $v_1(n) = 1$ has only finitely many solutions. The problem is Problem #889 of the Erdős problems database [EP889], and it is formalised in the Formal Conjectures repository [FC], where the variant `erdos_889.variants.v1_eq_1_finite` asks whether $\{n : v_1(n) = 1\}$ is finite.

Erdős and Selfridge also introduced a variant $V(n; k)$, the number of primes $p \mid n + k$ whose exact prime-power component $p^\alpha \parallel n + k$ divides none of $n, \dots, n + k - 1$, and asked the same question for $V_1(n) = 1$ (perhaps with largest solution 80). Kitamura [Ki26] has given a Lean 4 proof that $\{n : V_1(n) = 1\}$ is finite (repository dated 22 September 2026); its documentation states that it does not address the lowercase variant. Since $p \nmid n + i$ implies $p^\alpha \nmid n + i$, one has $v(n; k) \leq V(n; k)$, so $V_1(n) \geq v_1(n)$ and finiteness of $\{n : v_1(n) = 1\}$ implies finiteness of $\{n : V_1(n) = 1\}$, not conversely. (In particular, Theorem A below also shows that every solution of $V_1(n) = 1$ is smaller than N_0 .)

What this note contains. An explicit threshold N_0 with $\log \log N_0 = 45.28$ beyond which $v(n, k) \geq 2$ for some $k \leq 10 \log n$ (Theorem A), and an explicit version uniform in l . An elementary lemma (Lemma 3.1) shows that if $v(n, k) \leq 1$ for all $k \leq y$, then two of $n + 1, \dots, n + y$ have the form Wq^e with distinct primes q and small cofactors W ; Matveev's bound [Ma00] applied to their quotient gives $\log n < K(\log \log n)^5$ with an explicit K .

What is not new. The qualitative statement that $\{n : v_1(n) = 1\}$ is finite, with an effective (but not computed) bound, is a corollary of a published theorem: Langevin’s 1981 bound for the number of k -smooth integers among $n + 1, \dots, n + k$ [La81, Corollaire 1, (8)], combined with a ten-line elementary lemma. We write out this deduction in Section 7.1; the same route also gives the qualitative uniform-in- l statement. The method is also classical: Baker-type lower bounds applied to smooth numbers in blocks of consecutive integers go back to Tijdeman [Ti73], Ramachandra, Shorey and Tijdeman [RST75, RST76] and Langevin [La81], and the device of assigning each prime to the term in which it occurs to the highest power is of Sylvester–Erdős type. What this note adds is quantitative: an explicit threshold $\log \log N_0 = 45.28$ and an explicit uniform-in- l threshold 48.82. The threshold is far beyond any computation, so the conjecture that 330 is the largest solution of $v_1(n) = 1$ remains open, as does the main problem $v_0(n) \rightarrow \infty$.

2 Notation and statements

Throughout, $\log$ is the natural logarithm and p, q denote primes. For an integer $m \geq 2$, $P(m)$ is its largest prime factor, and $P(1) = 1$; $\omega(m)$ is the number of distinct prime factors of m ; v_p is the p -adic valuation; and $\pi(x) = \#\{p \leq x\}$ for real x . Following [ErSe67] and [FC], for integers $n \geq 0$, $k \geq 1$ and $l \geq 1$ put

$$v(n, k) = \#\{p : p \mid n + k, p \nmid n + i \text{ for all } 0 \leq i \leq k - 1\}, \quad v_l(n) = \sup_{k \geq l} v(n, k) \in \mathbb{N} \cup \{\infty\}.$$

Lemma 2.1. *For integers $n \geq 0$ and $k \geq 1$, $v(n, k) = \#\{p : p \mid n + k, p > k\}$.*

Proof. Let $p \mid n + k$. If $p \leq k$, then $i := k - p$ satisfies $0 \leq i \leq k - 1$ and $p \mid (n + k) - p = n + i$, so p is not counted. If $p > k$ and $0 \leq i \leq k - 1$, then $(n + k) - (n + i) = k - i \in [1, k]$ is not divisible by p , so $p \nmid n + i$ and p is counted. Thus $v(n, k) \leq 1$ means that $n + k$ has at most one prime factor greater than k . $\square$

Theorem A. *Put $\ell_1 := 45.28$ and $N_0 := \exp(\exp(\ell_1))$, so that $\log N_0 = e^{45.28} = 4.6223 \cdot 10^{19}$ (to five figures). For every integer $n \geq N_0$ there is an integer k with $1 \leq k \leq 10 \log n$ and $v(n, k) \geq 2$.*

Corollary A1. *Every n with $v_1(n) = 1$ satisfies $n < N_0$; in particular $\{n : v_1(n) = 1\}$ is finite.*

Proof. If $n \geq N_0$, Theorem A gives $k \geq 1$ with $v(n, k) \geq 2$, so $v_1(n) \geq 2$ and $v_1(n) \neq 1$. $\square$

In the terms of [FC], Corollary A1 is a paper proof, not a Lean proof, that `Erdos889.erdos_889.variants.v1_eq_1_finite` holds with answer `True`. It depends on the published results of Matveev, Rosser–Schoenfeld and Robin (Sections 4–5). The qualitative finiteness also follows from Langevin [La81] with Lemma 3.1(i) (Section 7.1).

Theorem A_l (fixed l). *Let $1 \leq l \leq 20$ and let $\ell_1(l)$ be as in Table 2 (so $\ell_1(1) = 45.28$, $\ell_1(2) = 45.35$, $\dots$, $\ell_1(20) = 45.58$). Every integer n with $\log \log n \geq \ell_1(l)$ has an integer k with $l \leq k \leq 10 l \log n$ and $v(n, k) \geq 2$.*

Theorem A_{unif} (uniform in l). *Let $l \geq 1$ and n be integers with $\log \log n \geq 48.82$ and $\log n \geq l$. Then there is an integer k with $l \leq k \leq 10 l \log n$ and $v(n, k) \geq 2$. Equivalently, this holds for every $n \geq N(l)$, where*

$$N(l) := \max(\exp(e^{48.82}), \lceil e^l \rceil), \quad e^{48.82} = 1.593 \cdot 10^{21}.$$

Corollary A_l. *For every $l \geq 1$ and every $n \geq N(l)$, $v_l(n) \geq 2$.*

Corollary A_l is the level-2 case of `erdos_889.variants.general` in [FC]; that variant asks for $v_l(n) \rightarrow \infty$, which is not proved here. Section 7.4 lists other questions that this note does not address.

3 The elementary reduction

Setting. Fix integers $l \geq 1$ and $n \geq 2$ and a real number y with $Y := \lfloor y \rfloor \geq l + 2$. Put $\Pi_l(n) := n(n + 1) \cdots (n + l - 1)$ and

$$\omega_{l,y}(n) := \#\{p \leq y : p \mid \Pi_l(n)\}, \quad \mathcal{P} := \{p \leq y : p \nmid \Pi_l(n)\}, \quad |\mathcal{P}| = \pi(y) - \omega_{l,y}(n).$$

The hypothesis is

$$v(n, k) \leq 1 \quad \text{for every integer } k \text{ with } l \leq k \leq y. \quad (\text{H})$$

Define

$$A := \pi(y) - 2\omega_{l,y}(n) - \frac{2\log((Y-l-1)!)}{\log n}, \quad \text{and, when } A > 1, \quad T := \frac{\log((Y-l-1)!)}{A-1}.$$

Lemma 3.1 (Lemma R). *Assume (H).*

(i) *For $p \in \mathcal{P}$ let $k_0(p) := \min\{k \geq l : p \mid n+k\}$. Then $l \leq k_0(p) \leq p-1 \leq Y-1$ and $P(n+k_0(p)) = p$. The map $p \mapsto k_0(p)$ is injective. Hence $J := k_0(\mathcal{P}) \subseteq \{l, l+1, \dots, Y-1\}$ has $|J| = \pi(y) - \omega_{l,y}(n)$, and for $j \in J$ every prime factor of $n+j$ is at most Y .*

(ii) *Assume $J \neq \emptyset$. For every prime $p \leq Y$ fix $j_p \in J$ with $v_p(n+j_p) = \max_{j \in J} v_p(n+j)$. For $j \in J$ let*

$$U_j := \prod_{p \leq Y, j_p=j} p^{v_p(n+j)}, \quad W_j := \prod_{p \leq Y, j_p \neq j} p^{v_p(n+j)}.$$

Then $n+j = U_j W_j$ for each $j \in J$, and $\prod_{j \in J} W_j \leq (Y-l-1)!$.

(iii) *Let $s(j) := \omega(U_j)$, $J_0 := \{j \in J : s(j) = 0\}$ and $J_1 := \{j \in J : s(j) = 1\}$. Then $|J_0| \leq \log((Y-l-1)!)/\log n$ and $|J_1| \geq \pi(y) - 2\omega_{l,y}(n) - 2|J_0| \geq A$.*

(iv) *For $j \in J_1$, $U_j = q_j^{e_j}$ with q_j a prime $\leq Y$ and $e_j \geq 1$. So $n+j = W_j q_j^{e_j}$ with $q_j \nmid W_j$. The map $j \mapsto q_j$ is injective on J_1 .*

(v) *If $A > 1$, then $J \neq \emptyset$, and there are $j_1 \neq j_2$ in J_1 with $\log W_{j_1} \leq T$ and $\log W_{j_2} \leq T$.*

Proof. (i) Let $p \in \mathcal{P}$. Among the p consecutive integers $n, \dots, n+p-1$ there is a multiple $n+i$ of p , with $0 \leq i \leq p-1$. If $p \leq l$, this $n+i$ divides $\Pi_l(n)$, contradicting $p \nmid \Pi_l(n)$; so $p \geq l+1$. Since $p \nmid n+i$ for $i < l$, the multiple has $l \leq i \leq p-1$; hence $l \leq k_0(p) \leq p-1$. As $p \leq y$ is an integer, $p \leq Y$, so $k_0(p) \leq Y-1 < y$ and (H) gives $v(n, k_0(p)) \leq 1$. By Lemma 2.1 (note $k_0(p) \geq 1$), $n+k_0(p)$ has at most one prime factor greater than $k_0(p)$. The prime p is such a factor, so every other prime factor of $n+k_0(p)$ is at most $k_0(p) < p$, and $P(n+k_0(p)) = p \leq Y$. If $k_0(p) = k_0(p')$, then $p = P(n+k_0(p)) = p'$; so the map is injective and $|J| = |\mathcal{P}|$.

(ii) By (i), $n+j = \prod_{p \leq Y} p^{v_p(n+j)}$ for $j \in J$; splitting the primes $p \leq Y$ according to whether $j_p = j$ gives $n+j = U_j W_j$. Fix a prime $p \leq Y$ and $j \in J$ with $j \neq j_p$. Since $v_p(n+j) \leq v_p(n+j_p)$, the power $p^{v_p(n+j)}$ divides both $n+j$ and $n+j_p$, hence divides $j-j_p \neq 0$, and $v_p(n+j) \leq v_p(|j-j_p|)$. Summing over $j \in J \setminus \{j_p\}$, a subset of $\{l, \dots, Y-1\} \setminus \{j_p\}$, and using that all terms are nonnegative,

$$\sum_{j \in J, j \neq j_p} v_p(n+j) \leq \sum_{\substack{i=l \\ i \neq j_p}}^{Y-1} v_p(|i-j_p|) = v_p((j_p-l)!(Y-1-j_p)!) \leq v_p((Y-l-1)!),$$

because $a!b!$ divides $(a+b)!$ and here $a+b = Y-l-1$. Therefore

$$\prod_{j \in J} W_j = \prod_{p \leq Y} p^{\sum_{j \in J, j \neq j_p} v_p(n+j)} \leq \prod_{p \leq Y} p^{v_p((Y-l-1)!)} = (Y-l-1)!,$$

the last equality because every prime factor of $(Y-l-1)!$ is at most Y .

(iii) For $j \in J_0$ we have $U_j = 1$, so $W_j = n+j > n$. All $W_j \geq 1$, so by (ii) $|J_0| \log n \leq \sum_{j \in J_0} \log W_j \leq \sum_{j \in J} \log W_j \leq \log((Y-l-1)!)$. Next, $s(j) = \#\{p \leq Y : j_p = j, v_p(n+j) \geq 1\}$. Every prime $p \leq Y$ has exactly one j_p , so these sets are pairwise disjoint subsets of the primes up to Y , and $\sum_{j \in J} s(j) \leq \pi(Y) = \pi(y)$. With $J_{\geq 2} := J \setminus (J_0 \cup J_1)$ this gives $|J_1| + 2|J_{\geq 2}| \leq \pi(y)$, while $|J_0| + |J_1| + |J_{\geq 2}| = |J| = \pi(y) - \omega_{l,y}(n)$. Eliminating $|J_{\geq 2}|$ gives $|J_1| \geq \pi(y) - 2\omega_{l,y}(n) - 2|J_0|$, which is at least A by the bound for $|J_0|$.

(iv) If $s(j) = 1$, there is exactly one prime $q = q_j \leq Y$ with $j_q = j$ and $v_q(n+j) \geq 1$; the other primes p with $j_p = j$ contribute $p^0 = 1$. So $U_j = q_j^{e_j}$ with $e_j := v_{q_j}(n+j) \geq 1$, and $q_j \nmid W_j$ because W_j contains only primes p with $j_p \neq j$. Finally q_j determines $j = j_{q_j}$.

(v) If $J = \emptyset$, then $\omega_{l,y}(n) = \pi(y)$ and $A \leq -\pi(y) < 1$; so $A > 1$ forces $J \neq \emptyset$. By (iii), $r := |J_1| \geq A > 1$, so the integer r is at least 2. List the numbers $\log W_j$ ($j \in J_1$) as $w_1 \leq w_2 \leq \dots \leq w_r$ and let j_1, j_2 be indices giving w_1, w_2 . All $w_i \geq 0$, so by (ii) $(r-1)w_2 \leq w_2 + \dots + w_r \leq \sum_{j \in J} \log W_j \leq \log((Y-l-1)!)$. Since $r-1 \geq A-1 > 0$, we get $w_1 \leq w_2 \leq T$. $\square$

4 The Baker step

4.1 Matveev's theorem

We use Corollary 2.3 of Matveev [Ma00] (Russian original p. 127, English translation p. 1219), in his notation and equation numbering, except that his number n of logarithms is called t here. Let $\mathbb{K} \subset \mathbb{C}$ be an algebraic number field of degree D over $\mathbb{Q}$; put $\varkappa = 1$ if $\mathbb{K} \subseteq \mathbb{R}$ and $\varkappa = 2$ otherwise. Let $\alpha_1, \dots, \alpha_t \in \mathbb{K}^*$ with absolute logarithmic heights $h(\alpha_j)$, let $\ln \alpha_1, \dots, \ln \alpha_t$ be arbitrary fixed *non-zero* values of the logarithms, let $b_1, \dots, b_t \in \mathbb{Z}$, and put

$$\Lambda = b_1 \ln \alpha_1 + \dots + b_t \ln \alpha_t \quad (1.1), \quad B = \max \left\{ 1, \max_{1 \leq j \leq t} \frac{|b_j| A_j}{A_t} \right\} \quad (1.3), \quad B^* = \max_{1 \leq j \leq t} |b_j| \quad (1.4),$$

$$A_j \geq \max \{ D h(\alpha_j), |\ln \alpha_j|, 0.16 \} \quad (2.4), \quad \Omega = A_1 \cdots A_t.$$

Theorem 4.1 (Matveev [Ma00, Corollary 2.3]). *If $\Lambda \neq 0$, the A_j satisfy (2.4) and B is given by (1.3), then*

$$\ln |\Lambda| > -C_1(t) D^2 \Omega \ln(eD) \ln(eB), \quad C_1(t) = C_1(t, \varkappa) = \min \left\{ \frac{1}{\varkappa} \left(\frac{1}{2} e t \right)^\varkappa 30^{t+3} t^{3.5}, 2^{6t+20} \right\},$$

where B may be replaced by B^* .

We use it only with $\mathbb{K} = \mathbb{Q} \subset \mathbb{R}$, so $D = 1$, $\varkappa = 1$ and $\ln(eD) = 1$, with the real logarithms. For a rational number a/b in lowest terms, $h(a/b) = \log \max(|a|, |b|)$. The constants are

$$C_1(3) = \min \left\{ \frac{3e}{2} 30^6 3^{3.5}, 2^{38} \right\} = \frac{3e}{2} 30^6 3^{3.5} = 1.39007 \dots \cdot 10^{11},$$

$$C_1(2) = \min \{ e 30^5 2^{3.5}, 2^{32} \} = e 30^5 2^{3.5} = 7.47318 \dots \cdot 10^8.$$

The frequently quoted simplified form $1.4 \cdot 30^{t+3} t^{4.5} D^2 (1 + \log D) (1 + \log B) A_1 \cdots A_t$ is weaker, since $1.4 \cdot 30^6 \cdot 3^{4.5} = 1.4319 \cdot 10^{11} \geq C_1(3)$; using it would give $\ell_1 = 45.31$ in place of 45.28.

4.2 Two smooth terms

Proposition 4.2. *In the setting of Section 3, assume (H), $A > 1$ and $Y \leq n$, and put $L := \log n$. Then*

$$L - \log Y < C_1(3) (\log Y)^2 \max(T, 0.16) (1 + \log \bar{B}), \quad \text{where } \bar{B} := \frac{\log(n+Y)}{\log 2}. \quad (4.1)$$

Proof. Take $j_1 \neq j_2$ from Lemma 3.1(v) and write $W_i := W_{j_i}$, $q_i := q_{j_i}$, $e_i := e_{j_i}$. Then $n + j_i = W_i q_i^{e_i}$, $q_1 \neq q_2$ (Lemma 3.1(iv)), $q_i \leq Y$, $e_i \geq 1$ and $\log W_i \leq T$ for $i = 1, 2$. Define, with real logarithms,

$$\Lambda := \log(n + j_1) - \log(n + j_2) = e_1 \log q_1 - e_2 \log q_2 + \log(W_1/W_2).$$

Then $\Lambda \neq 0$ because $j_1 \neq j_2$. By the mean value theorem $|\Lambda| = |j_1 - j_2|/\xi$ for some ξ between $n + j_1$ and $n + j_2$, so $\xi > n$; and $|j_1 - j_2| \leq Y - 1 - l < Y$. Hence $|\Lambda| < Y/n$, that is,

$$-\log |\Lambda| > L - \log Y. \quad (4.2)$$

Since $q_i^{e_i} \leq n + j_i < n + Y$, we have $e_i \leq \log(n + Y)/\log q_i \leq \bar{B}$, so $B^* := \max(e_1, e_2, 1) \leq \bar{B}$; note $\bar{B} \geq 1$.

Case 1: $W_1 \neq W_2$. Apply Theorem 4.1 with $t = 3$, $\alpha_1 = q_1$, $\alpha_2 = q_2$, $\alpha_3 = W_1/W_2$, $b_1 = e_1$, $b_2 = -e_2$, $b_3 = 1$. The logarithms are non-zero since $q_i \geq 2$ and $W_1/W_2 \neq 1$. We have $h(q_i) = \log q_i$. Writing $W_1/W_2 = a/b$ in lowest terms, $a \mid W_1$ and $b \mid W_2$, so $h(W_1/W_2) \leq \log \max(W_1, W_2) \leq T$; also $|\log(W_1/W_2)| \leq \max(\log W_1, \log W_2) \leq T$ because both logarithms are nonnegative. So $A_1 := \log q_1$, $A_2 := \log q_2$ and $A_3 := \max(T, 0.16)$ satisfy (2.4) ($\log q_i \geq \log 2 > 0.16$). Since $\Lambda \neq 0$, Theorem 4.1 with B^* in place of B gives

$$-\log |\Lambda| < C_1(3) \log q_1 \log q_2 \max(T, 0.16) (1 + \log B^*) \leq C_1(3) (\log Y)^2 \max(T, 0.16) (1 + \log \bar{B}).$$

Case 2: $W_1 = W_2$. Then $\ln \alpha_3 = 0$ is not an admissible value, and we use two logarithms instead: $\Lambda = e_1 \log q_1 - e_2 \log q_2 \neq 0$. Theorem 4.1 with $t = 2$, $\alpha_i = q_i$, $A_i = \log q_i$ gives

$$-\log |\Lambda| < C_1(2) \log q_1 \log q_2 (1 + \log B^*) \leq 0.16 C_1(3) (\log Y)^2 (1 + \log \bar{B}),$$

using $C_1(2) = 7.47 \cdot 10^8 \leq 0.16 C_1(3) = 2.22 \cdot 10^{10}$. This is at most the bound of Case 1.

In both cases, combining with (4.2) gives (4.1). $\square$

5 The numerical chain and N_0

Besides Theorem 4.1 we use two explicit estimates: $x/\log x < \pi(x)$ for real $x \geq 17$ (Rosser–Schoenfeld [RS62, Corollary 1, (3.5)]), and $\omega(m) \leq R \log m / \log \log m$ for $m \geq 3$, where $R := 1.3841$ (Robin [Ro83, Théorème 11]).

5.1 Constants

Fix an integer $l \geq 1$ and put $c := 10$. For real ℓ_1 define

$$\begin{aligned} m &:= \log(cl), & \kappa_1 &:= 1 + \frac{m}{\ell_1}, & \kappa_2 &:= 1 + \frac{1 - \log \log 2 + e^{-\ell_1}}{\ell_1}, \\ \mu &:= \frac{2Rm}{c - 2R}, & \rho &:= \frac{\ell_1 + m - 1}{\ell_1 - \mu}, & \varepsilon_1 &:= \frac{3e^{-\ell_1}}{c(\ell_1 + m - 1)}, \\ \beta &:= 2c + \frac{2c(m - 1) + 2R/\ell_1 + 6e^{-\ell_1} + 1}{\ell_1}, & \varepsilon_2 &:= \frac{\beta \kappa_1 \ell_1^2 e^{-\ell_1}}{(c - 2R)(1 - \mu/\ell_1)}, & \kappa_3 &:= \frac{c}{c - 2R} \rho \kappa_1 \frac{1 + \varepsilon_1}{1 - \varepsilon_2}, \\ K &:= C_1(3) \kappa_1^2 \kappa_2 \kappa_3 + \frac{\kappa_1}{\ell_1^4}, & \varphi(\ell_1) &:= \ell_1 - 5 \log \ell_1 - \log K. \end{aligned} \tag{5.1}$$

Here $-\log \log 2 = 0.36651\dots$. Note that $m \geq \log 10 > 1$, and that $\kappa_3 > 1$ whenever $0 < \varepsilon_2 < 1$ and $\ell_1 > \mu$.

Proposition 5.1. *Let $l \geq 1$, and let $\ell_1 \geq 10$ satisfy $\ell_1 > \mu$, $\varepsilon_2 < 1$ and $\varphi(\ell_1) \geq 0$. Then every integer n with $\log \log n \geq \ell_1$ and $\log n \geq l$ has an integer k with $l \leq k \leq 10l \log n$ and $v(n, k) \geq 2$.*

Proof. Let $L := \log n$, $\ell := \log L \geq \ell_1 \geq 10$, $y := clL$ and $Y := \lfloor y \rfloor$. Suppose, for contradiction, that (H) holds: $v(n, k) \leq 1$ for every integer $k \in [l, y]$. Then $L \geq e^{10}$, so $y \geq 10L \geq 67$; $Y > y - 1 \geq l + 2$; $Y \leq y \leq cL^2 < e^L = n$ (as $l \leq L$); $n \geq 26$ and $l - 1 < n$; and $\log y = \ell + m$.

S1 (primes). By [RS62, (3.5)] with $x = y$: $\pi(y) > y/\log y = clL/(\ell + m)$.

S2 (ω). Every prime dividing $\Pi_l(n)$ divides some $n + i$ with $0 \leq i \leq l - 1$, and $n + i \geq 3$, so by [Ro83]

$$\omega_{l,y}(n) \leq \sum_{i=0}^{l-1} \omega(n + i) \leq \sum_{i=0}^{l-1} \frac{R \log(n + i)}{\log \log(n + i)} \leq \frac{lR(L + 1)}{\ell},$$

using $\log(n + i) \leq L + (l - 1)/n \leq L + 1$ and $\log \log(n + i) \geq \log \log n = \ell > 0$.

S3 (factorial). Let $G(x) := (x + 1) \log x - x + 1$. For integers $M \geq 1$, $\log M! = \sum_{i=1}^{M-1} \log i + \log M \leq \int_1^M \log t \, dt + \log M = G(M)$, and G is increasing on $[1, \infty)$ since $G'(x) = \log x + 1/x > 0$. With $M = Y - l - 1 \in [1, y - l - 1]$,

$$\log((Y - l - 1)!) \leq G(y - l - 1) = (y - l) \log(y - l - 1) - (y - l - 1) + 1 \leq y \log y - y + l + 2 = lLc(\ell + m - 1) + l + 2.$$

S4 (A). By S1–S3 and the definition of A (with $\log n = L$),

$$A - 1 \geq \frac{clL}{\ell + m} - \frac{2lRL}{\ell} - \frac{2lR}{\ell} - 2cl(\ell + m - 1) - \frac{2(l + 2)}{L} - 1 = lLa(\ell) - E(\ell),$$

where

$$a(\ell) := \frac{c}{\ell + m} - \frac{2R}{\ell} = \frac{(c - 2R)(\ell - \mu)}{\ell(\ell + m)}, \quad E(\ell) := \frac{2lR}{\ell} + 2cl(\ell + m - 1) + \frac{2(l + 2)}{L} + 1.$$

Since $\ell \geq \ell_1 > \mu$, $a(\ell) > 0$. Using $1/l \leq 1$, $(l + 2)/l \leq 3$, $1/L = e^{-\ell} \leq e^{-\ell_1}$, $1/\ell \leq 1/\ell_1$ and $m - 1 > 0$,

$$\frac{E(\ell)}{l} \leq 2cl + \left(2c(m - 1) + \frac{2R}{\ell_1} + 6e^{-\ell_1} + 1\right) \leq \beta\ell, \quad \frac{1}{a(\ell)} = \frac{\ell + m}{(c - 2R)(1 - \mu/\ell)} \leq \frac{\kappa_1 \ell}{(c - 2R)(1 - \mu/\ell_1)}.$$

Hence $E(\ell)/(lLa(\ell)) \leq [\beta\kappa_1/((c - 2R)(1 - \mu/\ell_1))] \ell^2 e^{-\ell} \leq \varepsilon_2$, because $\ell^2 e^{-\ell}$ is decreasing for $\ell \geq 2$. Therefore $A - 1 \geq lLa(\ell)(1 - \varepsilon_2) > 0$.

S5 (T). By S3, $\log((Y-l-1)!) \leq lLc(\ell+m-1)(1+\varepsilon_1)$, since $(l+2)/(lLc(\ell+m-1)) \leq 3e^{-\ell_1}/(c(\ell_1+m-1)) = \varepsilon_1$. With S4,

$$T \leq \frac{c(\ell+m-1)(1+\varepsilon_1)}{a(\ell)(1-\varepsilon_2)} = \frac{c}{c-2R} \cdot \frac{\ell+m-1}{\ell-\mu} \cdot \ell(\ell+m) \cdot \frac{1+\varepsilon_1}{1-\varepsilon_2} \leq \kappa_3 \ell^2,$$

because $(\ell+m-1)/(\ell-\mu)$ is decreasing in ℓ (its derivative has the sign of $-(m-1+\mu) < 0$), hence at most ρ , and $\ell+m \leq \kappa_1 \ell$. Also $0.16 \leq \kappa_3 \ell^2$, so $\max(T, 0.16) \leq \kappa_3 \ell^2$.

S6 ($\bar{B}$ and $\log Y$). We have $\log(n+Y) \leq L+Y/n \leq L+1$ and $\log(L+1) \leq \ell+1/L$, so

$$1 + \log \bar{B} \leq 1 + \ell + e^{-\ell_1} - \log \log 2 \leq \kappa_2 \ell, \quad \log Y \leq \log y = \ell + m \leq \kappa_1 \ell.$$

S7 (conclusion). All hypotheses of Proposition 4.2 hold ($A > 1$ by S4, and $Y \leq n$). By (4.1) and S5–S6,

$$L < \kappa_1 \ell + C_1(3) \kappa_1^2 \ell^2 \cdot \kappa_3 \ell^2 \cdot \kappa_2 \ell \leq K \ell^5,$$

using $\kappa_1 \ell \leq (\kappa_1/\ell_1^4) \ell^5$. That is, $\log n < K(\log \log n)^5$. But $\psi(\ell) := \ell - 5 \log \ell$ has $\psi'(\ell) = 1 - 5/\ell > 0$ for $\ell > 5$, so $\psi(\ell) \geq \psi(\ell_1) \geq \log K$, which means $L = e^\ell \geq K \ell^5$. This contradiction proves the proposition. $\square$

5.2 Proof of Theorem A

For $l = 1$ and $\ell_1 = 45.28$ the script `n0_compute.py` evaluates every quantity in (5.1) as an interval (mpmath interval arithmetic, 60 digits, outward rounding) and certifies from the interval endpoints every numerical inequality used above: $\varphi(\ell_1) \geq 0$, $\ell_1 \geq 10$, $\ell_1 > \mu$, $\varepsilon_2 < 1$, $\kappa_3 \ell_1^2 \geq 0.16$, $m > 1$, $y \geq 67$ and $C_1(2) \leq 0.16 C_1(3)$. The decimal inputs 1.3841 and 45.28 enter as enclosing intervals. The monotonicity facts are proved in the text. Table 1 lists the values, rounded for display.

quantity	value	script name	quantity	value	script name
$m = \log 10$	2.30258509299	m	β	20.5987817363	e1
κ_1	1.05085214428	kappa1	ε_2	$1.35404300089 \cdot 10^{-16}$	eps2
κ_2	1.03017917227	kappa2	κ_3	1.52457710226	kappa3
μ	0.881387214031	mu	$C_1(3)$	$1.39007316922 \cdot 10^{11}$	C1(3)
ρ	1.04919010235	ratio	K	$2.41092138023 \cdot 10^{11}$	K
ε_1	$1.39329644231 \cdot 10^{-22}$	eps1	$\varphi(45.28)$	$0.00722781770539 > 0$	phi

Table 1: The constants (5.1) for $l = 1$, $c = 10$, $\ell_1 = 45.28$.

So Proposition 5.1 applies with $l = 1$; the condition $\log n \geq 1$ is automatic. Every n with $\log \log n \geq 45.28$, that is every $n \geq N_0 = \exp(e^{45.28})$, has some k with $1 \leq k \leq 10 \log n$ and $v(n, k) \geq 2$. This proves Theorem A. $\square$

The value 45.28 is the least multiple of 0.01 passing the certificate; 45.27 fails. The simplifications in S4–S7 cost less than 0.01 in ℓ_1 : evaluated directly from S1–S3 and S6, the right-hand side of (4.1) first falls below L near $\ell \approx 45.272$ (not used), and it is at most $K \ell^5$ at $\ell \in \{45.28, 46, 50, 60, 100, 1000\}$, a check of the algebra in S4–S7.

6 The version uniform in l

For fixed l , Proposition 5.1 with the constants (5.1) gives the thresholds in Table 2, which proves Theorem A_l . They depend on l only through $m = \log(10l)$ and grow slowly: $\ell_1(l) = 45.95, 46.53, 47.54$ at $l = 10^3, 10^6, 10^{12}$. For all $1 \leq l \leq \log n$ at once, Proposition 6.1 gives the single threshold $\log \log n \geq 48.82$.

For the uniform version put $c = 10$, $R = 1.3841$, let κ_2 be as in (5.1), and define

$$\begin{aligned} \kappa_1^u &:= 2 + \frac{\log c}{\ell_1}, & \mu^u &:= \frac{2R \log c}{c - 4R}, & \rho^u &:= \frac{2\ell_1 + \log c - 1}{\ell_1 - \mu^u}, & \varepsilon_1^u &:= \frac{3e^{-\ell_1}}{c(\ell_1 + \log c - 1)}, \\ \beta^u &:= 4c + \frac{2c(\log c - 1) + 2R/\ell_1 + 6e^{-\ell_1} + 1}{\ell_1}, & \varepsilon_2^u &:= \frac{\beta^u \kappa_1^u \ell_1^2 e^{-\ell_1}}{(c - 4R)(1 - \mu^u/\ell_1)}, \\ \kappa_3^u &:= \frac{c}{c - 4R} \rho^u \kappa_1^u \frac{1 + \varepsilon_1^u}{1 - \varepsilon_2^u}, & K^u &:= C_1(3)(\kappa_1^u)^2 \kappa_2 \kappa_3^u + \kappa_1^u/\ell_1^4, & \varphi^u(\ell_1) &:= \ell_1 - 5 \log \ell_1 - \log K^u. \end{aligned} \tag{6.1}$$

l	1	2	3	4	5	6	7	8	9	10
$\ell_1(l)$	45.28	45.35	45.39	45.42	45.44	45.46	45.48	45.49	45.50	45.51
l	11	12	13	14	15	16	17	18	19	20
$\ell_1(l)$	45.52	45.53	45.54	45.54	45.55	45.56	45.56	45.57	45.57	45.58

Table 2: Certified thresholds for fixed l ($c = 10$): $\ell_1(l)$ is the least multiple of 0.01 passing the certificate of Proposition 5.1. Correspondingly $\log N(l) = e^{\ell_1(l)}$ runs from $4.622 \cdot 10^{19}$ ($l = 1$) to $6.239 \cdot 10^{19}$ ($l = 20$), and $K(l)$ from $2.411 \cdot 10^{11}$ to $3.154 \cdot 10^{11}$.

Proposition 6.1. *Suppose $c > 4R$, $\ell_1 \geq 10$, $\ell_1 > \mu^u$, $\varepsilon_2^u < 1$ and $\varphi^u(\ell_1) \geq 0$. Then for every integer $l \geq 1$ and every integer n with $\log \log n \geq \ell_1$ and $\log n \geq l$, some k with $l \leq k \leq cl \log n$ has $v(n, k) \geq 2$.*

Proof. Repeat the proof of Proposition 5.1. Now $m = \log(cl)$ is not fixed; from $1 \leq l \leq L$ we only know $\log c \leq m \leq \ell + \log c$. The preliminary facts and S1–S3 are unchanged. The later steps change as follows.

S4. $a(\ell) = c/(\ell + m) - 2R/\ell$ is decreasing in m , so $a(\ell) \geq a^u(\ell) := c/(2\ell + \log c) - 2R/\ell = (c - 4R)(\ell - \mu^u)/(\ell(2\ell + \log c)) > 0$. Using $m \leq \ell + \log c$, $E(\ell)/l \leq 2R/\ell_1 + 2c(2\ell + \log c - 1) + 6e^{-\ell_1} + 1 \leq \beta^u \ell$ (as $\log c > 1$). Also $1/a^u(\ell) \leq \kappa_1^u \ell / ((c - 4R)(1 - \mu^u/\ell_1))$. Hence $E/(lLa) \leq E/(lLa^u) \leq \varepsilon_2^u$ and $A - 1 \geq lLa(\ell)(1 - \varepsilon_2^u) > 0$.

S5. Since $m \geq \log c$, $(l + 2)/(lLc(\ell + m - 1)) \leq \varepsilon_1^u$. Also $c(\ell + m - 1)/a(\ell) \leq c(2\ell + \log c - 1)/a^u(\ell) \leq \frac{c}{c-4R} \rho^u \kappa_1^u \ell^2$, because $(2\ell + \log c - 1)/(\ell - \mu^u)$ is decreasing (its derivative has the sign of $-(2\mu^u + \log c - 1) < 0$). So $T \leq \kappa_3^u \ell^2$, and $\kappa_3^u \ell_1^2 \geq 0.16$.

S6. $\log Y \leq \ell + m \leq 2\ell + \log c \leq \kappa_1^u \ell$. The bound $1 + \log \bar{B} \leq \kappa_2 \ell$ is unchanged; it uses $Y \leq cL^2 < n$.

S7. $L < K^u \ell^5$, which contradicts $\varphi^u(\ell_1) \geq 0$ as before. $\square$

With $c = 10$ (and $4R = 5.5364 < 10$) the script certifies $\ell_1 = 48.82$ (and 48.81 fails), with $\kappa_1^u = 2.04716$, $\mu^u = 1.42800$, $\rho^u = 2.08775$, $\beta^u = 40.5553$, $\varepsilon_2^u = 2.87 \cdot 10^{-17}$, $\kappa_3^u = 9.57515$, $K^u = 5.73427 \cdot 10^{12}$ and $\varphi^u(48.82) = 0.00182 > 0$; it also certifies $\ell_1 > \mu^u$, $\varepsilon_2^u < 1$, $\kappa_3^u \ell_1^2 \geq 0.16$ and $\log c > 1$. This proves Theorem A_{unif}: if $n \geq N(l)$ then $\log \log n \geq 48.82$ and $\log n \geq l$. Corollary A_l follows as Corollary A1 does. $\square$

7 Remarks

7.1 The Langevin route

The qualitative content of Corollary A1 follows from a theorem of Langevin and the case $l = 1$ of Lemma 3.1(i), restated here; this route was pointed out in the prior-art check for this note.

Lemma 7.1. *Let $n \geq 1$ and $K \geq 2$ be integers, and assume $v(n, k) \leq 1$ for every $1 \leq k \leq K$. For each prime $p \leq K$ with $p \nmid n$ let $k_0(p)$ be the least $k \geq 1$ with $p \mid n + k$. Then $1 \leq k_0(p) \leq p - 1$, $P(n + k_0(p)) = p$, and $p \mapsto k_0(p)$ is injective. Hence $\#\{1 \leq k \leq K : P(n + k) \leq K\} \geq \pi(K) - \omega(n)$.*

Proof. Since $p \nmid n$, one of $n + 1, \dots, n + p - 1$ is divisible by p , so $1 \leq k_0(p) \leq p - 1 < p$ and p is a prime factor of $n + k_0(p)$ greater than $k_0(p)$. As $v(n, k_0(p)) \leq 1$, Lemma 2.1 shows that p is the only such factor; every other prime factor of $n + k_0(p)$ is at most $k_0(p) < p$, so $P(n + k_0(p)) = p \leq K$. This recovers p from $k_0(p)$, so the map is injective, and the $\pi(K) - \omega(n)$ primes $p \leq K$ with $p \nmid n$ give that many distinct $k \in [1, K]$ with $P(n + k) \leq K$. $\square$

Langevin [La81, Corollaire 1] considers integers n, k, a with $n > 1$, $a \geq 1$, $(n, a) = 1$, $k \geq 2$ and $a < n^{t_1}$ (his condition (1)). His constants C and c are effectively computable in terms of the parameters $t_1, \dots, t_6$ of his Théorème 1, which allows any real $t_1 < 1$; his Exemple (p. 242) takes $t_1 = \frac{1}{2}$, $t_2 = 1$, $t_3 = \frac{1}{3}$, $t_4 = 3$, $t_5 = t_6 = 6$. Part (8) of Corollaire 1 states: for every $\varepsilon > 0$ there is an explicit $r(\varepsilon)$, depending on C , c and ε , such that if $\log n / \log k \geq r(\varepsilon)$ then

$$\text{card } \mathcal{Y} < \max\left(2, (1 + c + \varepsilon) \pi(k) \frac{\log \log X}{\log X}\right), \quad X := \frac{\log n}{(\log k)^2},$$

where $\mathcal{Y}$ is the set of the numbers $n + ia$ ($1 \leq i \leq k$) with $P(n + ia) \leq k$.

Proposition 7.2. *Assuming [La81, Corollaire 1, (8)], there is an effectively computable N_1 such that $v_1(n) \geq 2$ for all $n \geq N_1$; in particular $\{n : v_1(n) = 1\}$ is finite.*

Proof. Fix Langevin's example parameters (so $t_1 = \frac{1}{2}$) and $\varepsilon = 1$. For large n put $k^* := \lfloor 10 \log n \rfloor$ and $a = 1$; condition (1) holds since $1 < n^{1/2}$. As $n \rightarrow \infty$, $\log n / \log k^* \rightarrow \infty$ and $X = \log n / (\log k^*)^2 \rightarrow \infty$, so for large n (8) applies and gives $\text{card } \mathcal{Y} < \max(2, o(\pi(k^*)))$. Suppose $v(n, k) \leq 1$ for every $1 \leq k \leq k^*$. Lemma 7.1 with $K = k^*$ gives $\text{card } \mathcal{Y} \geq \pi(k^*) - \omega(n)$. By [Ro83], $\omega(n) \leq R \log n / \log \log n$, and by [RS62], $\pi(k^*) > k^* / \log k^*$ with $k^* > 10 \log n - 1$ and $\log k^* \leq \log \log n + \log 10$; hence $\omega(n) \leq (R/10 + o(1)) \pi(k^*)$ and $\text{card } \mathcal{Y} \geq (1 - 0.13841 - o(1)) \pi(k^*)$, which contradicts the upper bound for large n . So for all $n \geq N_1$ some $k \leq 10 \log n$ has $v(n, k) \geq 2$, and every constant involved is effective. $\square$

With Lemma 3.1(i) in place of Lemma 7.1 and $k^* := \lfloor 10 l \log n \rfloor$, the same argument gives Corollary A_l qualitatively, also uniformly for $1 \leq l \leq \log n$: the numbers $n+j$ ($j \in J$) lie in $\mathcal{Y}$, $\omega_{l, k^*}(n) \leq (2R/10 + o(1)) \pi(k^*)$ and $X \geq L/(2\ell + \log 10)^2 \rightarrow \infty$.

We have not refereed Langevin's proof; we rely on the statement as printed. With his example constants ($C = \exp 10^6$, $c = 2000$ when $k > \exp 10^4$) a rough estimate, which we have not verified, suggests that the condition $\log n / \log k^* \geq r(\varepsilon)$ alone forces $\log \log n$ to be at least of order 10^6 . By contrast, the bound of Ramachandra, Shorey and Tijdeman [RST76, Theorem 2], that at most $\pi(k)$ of $u+1, \dots, u+k$ are k -smooth when $u \geq \exp(C(\log k)^2)$, is exactly borderline against Lemma 7.1 and gives no contradiction by itself.

7.2 Sharper inputs and the constant 10

Three sharper inputs lower the threshold of Theorem A to $\log \log n \geq 44.96$, that is $\log N'_0 = e^{44.96} = 3.356 \cdot 10^{19}$, a factor 0.73 below $\log N_0$: in S1 the bound $\pi(y) > y/(\log y - \frac{1}{2})$ for $y \geq 67$ [RS62, (3.3)]; in S2 Robin's $\omega(m') \leq \log m' / (\log \log m' - 1.1714)$ for $m' \geq 26$ [Ro83, Théorème 13]; and in Case 1 of Proposition 4.2 Matveev's B from (1.3) in place of B^* , with $\alpha_3 = W_1/W_2$ ordered last and $A_3 := \kappa'_3 \ell^2 \geq T$. This gives $L < \kappa_1 \ell + C_1(3) \kappa_1^2 \kappa'_3 \ell^4 (\ell - d')$ with explicit constants ($\kappa'_3 = 1.39730$ and $d' = 6.94609$ at $\ell_1 = 44.96$). The function $Q(\ell) := (\kappa_1 \ell + C_1(3) \kappa_1^2 \kappa'_3 \ell^4 (\ell - d')) e^{-\ell}$ is non-increasing for $\ell \geq \ell_1$ (its log-derivative is at most $4/\ell + 1/(\ell - d') - 1 < 0$), and the script certifies $Q(44.96) = 0.99329 < 1$ with all side conditions, including that the two-logarithm case is dominated; 44.95 fails. The derivation is written out in the accompanying notes (PROOF_THEOREM_A.md, Section 7.2). We state Theorem A with the simpler 45.28.

The constant 10 is not essential. For fixed l the argument works with any $c > 2R = 2.7682$, since then $a(\ell) > 0$ for large ℓ , with a threshold that grows as c decreases towards $2R$; with Robin's Théorème 13 any $c > 2$ works. The uniform version as written needs $c > 4R$. We have not tried to optimise c or the thresholds.

7.3 Finite data

These computations are evidence only; none is used in the proofs. *Lemma 3.1.* The script's check mode verifies every conclusion of Lemma 3.1(i)–(v) on every triple (n, l, Y) with $3 \leq n \leq 10^6$, $l \in \{1, 2, 3\}$, $l+2 \leq Y \leq 150$ and $v(n, k) \leq 1$ for all $l \leq k \leq Y$: 91,709 triples, 40,312 of them with $|J_1| \geq 2$, and no failure. The referee's separate implementation in exact integer arithmetic (not distributed) checked 207,874 instances with $l \leq 4$ and $Y \leq 200$, also without failure.

Least witnesses up to 10^7 . For every $1 \leq n \leq 10^7$ the least $k \geq 1$ with $v(n, k) \geq 2$ was searched in the window $k \leq \lfloor 30 \log(n+2) + 30 \rfloor$, which contains $[1, 10 \log n]$. Exactly 29 values of n have no such k in the window, namely the 29 values of Erdős and Selfridge listed in the introduction; so for $2 \leq n \leq 10^7$ the conclusion of Theorem A fails exactly at the 28 values $2 \leq n \leq 330$ of that list. For every other $n \leq 10^7$ the least such k is at most 11, and $k = 11$ occurs only at $n = 210$ ($221 = 13 \cdot 17$) and $n = 840$ ($851 = 23 \cdot 37$).

Exception lists for $l \leq 20$. For $l = 1, 2, 3, 5, 8, 13, 20$ the numbers $n \leq 10^7$ with no $k \in [l, \lfloor 40 \log(n+2) + 40 \rfloor]$ satisfying $v(n, k) \geq 2$ number 29, 103, 187, 383, 722, 1531, 2817, with largest elements 330, 1365, 1365, 2415, 4895, 11655, 27714. This truncates $\sup_{k \geq l} v_l(n) \geq 2$, every n not listed has $v_l(n) \geq 2$, but a listed n is only known to have no witness in the window.

The witness computation was cross-checked against an independent brute-force program for $n \leq 2 \cdot 10^6$, and the two rows with least $k > 10$ were re-verified, including minimality of k , by an independently written trial-division checker sharing no code with the generator.

7.4 Open questions not addressed

This note does not address the main problem $v_0(n) \rightarrow \infty$, nor $v_l(n) \rightarrow \infty$ for fixed l ; nor $v_0(n) \geq 3$ or $v_l(n) \geq 3$ for all large n ; nor the conjectures that 330 is the largest n with $v_1(n) = 1$ and that 80 is the largest n with $V_1(n) = 1$. Since N_0 is far beyond computation, the set $\{n : v_1(n) = 1\}$ is still not determined.

Provenance and verification status

The proof in this note was found and written by an AI pipeline built on Claude models (Anthropic): Claude Opus for proof search and writing, an independent Claude referee working in a fresh context, and Claude Fable coordinating. The referee checked every step against the primary sources and recomputed the constants independently; its verdict was “pass with repairs”, the repairs concerned wording and citations only, and they are incorporated here. This is a same-vendor review; no cross-vendor or human referee has checked the proof. The numerical constants were certified by interval arithmetic (`n0_compute.py`, `mpmath`); the witness computation was checked by an independently written program. The author takes responsibility for the mathematics. There is no Lean formalisation; the proof uses Matveev’s theorem, which is not available in Mathlib. The literature search was done by the same pipeline, in two passes without general web search. The following sources could not be consulted: Erdős [Er98, p. 178]; Shorey and Tijdeman [ST86]; Erdős [Er76]; Tijdeman’s review in Math. Rev. 54 (1977), cited in [La81] for the case $a = 1$ of (8); and the Math. Rev. review of [La81]. Any of them may already contain Theorem A or its qualitative form. Code and data: <https://github.com/chy4pro/automath>, directory `problems/erdos889`.

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